

Open Science and Reproducibility in demography, population, and data science

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Max Planck Society Open Science
Ambassadors Meeting 2026 - Berlin

Universität
Rostock



Slides

tinyurl.com/OSA26AK



MAX PLANCK INSTITUTE
FOR DEMOGRAPHIC RESEARCH

Agenda

Examples of OS activities for:

- **Education** ==> Open course materials; OS Training events
- **Research** ==> OS best practices; Reproducible workflows
 - Open data ==> Enabling future research
- **Community** ==> Finding other Open Science enthusiasts

Disclaimer: This is by **no means** a comprehensive list of MPIDR's OS activities. These mention only some where I was involved; and to receive your feedback.

EDUCATION

Open course materials; OS Training events

Master's courses materials + final essay hand-over via Git, GitHub = 2 birds, 1 stone!

This screenshot shows the GitHub interface for the repository 'Course_introduction_to_computational_social_science'. The repository is public and managed by user 'akbaritabar'. It features a file browser on the left with a list of files: 'Hands_on', 'Slides_PDF', 'Students', 'ZZZ_Final_examination', '.gitignore', 'LICENSE', and 'README.md'. The 'README.md' file is selected and its content is displayed in the main area. The README describes the course materials and provides a folder structure: a root folder containing a 'ReadMe.md' text file and a '.gitignore' file. The right sidebar shows repository statistics, including 53 commits, 8 stars, 5 watchers, and 1 fork. It also lists the license as GPL-3.0 and shows the release history with the latest release being '2025 summer semester e...'.

Course Introduction to Computational Social Science 2025

Materials, slides, hands-on code and assignments for the course "Introduction to computational social science" for the 2025 edition

Here is the folder structure and what to expect:

- Root folder
 - This ReadMe.md text file
 - File named ".gitignore" is very important and described how it works in the slides of week on open-science and reproducibility

First week, on-boarding on how to use Git, GitHub!

This screenshot shows the GitHub interface for the repository 'Course_computational_approaches_to_migration_research'. The repository is public and managed by user 'akbaritabar'. It features a file browser on the left with a list of files: 'Hands_on', 'Slides_PDF', 'Students', 'ZZZ_Final_examination', '.gitignore', 'LICENSE', 'ReadMe.md', and 'TODO_before_the_next_session.png'. The 'README' file is selected and its content is displayed in the main area. The README describes the course materials and provides a folder structure: a root folder containing a 'ReadMe.md' text file. The right sidebar shows repository statistics, including 63 commits, 8 stars, 5 watchers, and 1 fork. It also lists the license as GPL-3.0 and shows the release history with the latest release being '2025-2026 winter semester...'.

Course Computational approaches to migration research, winter semester 2026-27

Materials, slides, hands-on code and assignments for the course "Computational approaches to migration research" for the 2026-27 edition

Here is the folder structure and what to expect:

- Root folder
 - This ReadMe.md text file

This screenshot shows the GitHub interface for the repository 'Course_introduction_to_demographic_methods'. The repository is public and managed by user 'akbaritabar'. It features a file browser on the left with a list of files: 'Hands_on', 'Slides_PDF', 'Students', 'ZZZ_Final_examination', '.gitignore', 'LICENSE', 'ReadMe.md', and 'TODO_before_the_next_session.png'. The 'README' file is selected and its content is displayed in the main area. The README describes the course materials and provides a folder structure: a root folder containing a 'ReadMe.md' text file. The right sidebar shows repository statistics, including 64 commits, 9 stars, 1 watcher, and 0 forks. It also lists the license as GPL-3.0 and shows the release history with the latest release being '2026 summer semester...'.

Course Introduction to Demographic Methods, summer semester 2026

Materials, slides, hands-on code and assignments for the course "Introduction to Demographic Methods" for the 2026 edition.

Here is the folder structure and what to expect:

Co-organized workshop at EPC 2026, Bologna

Open Science Practices for Demographic Research

Jonas Schöley

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 [@jschoeley.com](https://twitter.com/jschoeley)

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4 + 1 Layers

Layer 0: Aspirational

Layer 1: Personal

Layer 2: Communal

Layer 3: Computational

Layer 4: Institutional

Jonas Schöley
had a great set of materials
on layers 0-3

See here:
[https://github.com/
jschoeley/epc26os](https://github.com/jschoeley/epc26os)

4 + 1 Layers

Layer 0: Aspirational

Layer 1: Personal

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See here:
[https://github.com/
jschoeley/epc26os](https://github.com/jschoeley/epc26os)

Institutional open science



Aliakbar **Akbaritabar**



Open science at the Max Planck Society. osip.mpdl.mpg.de.

A few questions!

- Who **owns** the scripts/codes developed by researchers?
- Who should receive Open Science **training**?
 - At which **stage** of the **career**?
- Who should provide OS **infrastructure** and **maintain** it?
- What software solutions should be used and encouraged?
Proprietary?
- How should OS activities be **recognized/rewarded**?

What can institutions do for OS? (1/2)

- Permit and encourage researchers to **share** their research products under an **open license**.
- Particularly, researchers **with limited contracts** should have the right to access and publish all research products they developed during their time with the institute, even after they leave.
- **Train** researchers in OS practices, funded by the institute.
- Create a centralized, in-house **research data infrastructure**, and maintain it.
- **Binding guidelines** for the organization, storage, and retrieval of research data, ensuring continuity of access and management beyond the tenure of individual researchers or teams.

What can institutions do for OS? (2/2)

- Prioritize the use and development of **open-source research software**.
- Move away from **proprietary** software solutions.
- Micro-funds for **OS projects**, and/or **dedicated staff** to support OS.
- Transparent and reproducible research as a standard **criterion** for institute's **evaluation**.
- **Recognize** OS contributions by researchers and **reward** them, especially for **early career** researchers.

Hands-on part

What to expect in the Hands-on part?

- An interactive, self-paced, course on Git, GitHub, and version control with 4 steps:
 - Git and version control basics
 - GitHub - remote repositories and online collaboration
 - Branches, collaboration, and open science
 - Advanced Git, GitHub tools, and reproducibility
- **Requirement:**
 - Having a GitHub account, or opening one! Otherwise, follow on a neighbor's screen.
 - Pair up the participants: based on knowledge of Git/GitHub.
- **Link:**
 - <https://tinyurl.com/EPC26OS>



About 35 students have so far taken the self-paced Git, GitHub course!

Git-GitHub-Version-Control-Interactive-Self-paced-Course

Unpin Watch 0 Fork 35 Star 10

main 1 Branch 2 Tags

akbaritabar minor edits in step 2 ✓ ee851a3 · 4 months ago 25 Commits

.course_config/questions	Logic extended to have four steps instead of two and allo...	5 months ago
.github	Updated instructions on how to mark a step as done sinc...	4 months ago
.instructor_documents	Updating label used for new participants to remove unde...	4 months ago
content/steps	minor edits in step 2	4 months ago
images	course content updated to match my previous course slid...	5 months ago
.gitignore	course content updated to match my previous course slid...	5 months ago
FAQ.md	Updating label used for new participants to remove unde...	4 months ago
LICENSE	course content updated to match my previous course slid...	5 months ago
README.md	Updated instructions on how to mark a step as done sinc...	4 months ago

README GPL-3.0 license

Interactive Course on Basics of Git, GitHub, and Version Control

This repository contains a self-paced interactive course on basics of Git, GitHub, version control, and reproducible collaboration.

Instructor: Aliakbar Akbaritabar

Affiliation: Assistant Professor of Computational Social Science at the University of Rostock and Max Planck Institute for Demographic Research

Contact: <https://github.com/akbaritabar>

About

This repository contains a self-paced interactive course on basics of Git, GitHub, version control, and reproducible collaboration. No prior knowledge is assumed. The only requirement is to have a free GitHub account. You can Sign up here: <https://github.com/signup?>

[akbaritabar.github.io/Git-GitHub-V...](https://github.com/akbaritabar/Git-GitHub-Version-Control-Interactive-Self-paced-Course)

academic-writing git github gitignore open-science reproducibility reproducible-research workflows

Readme

GPL-3.0 license

Activity

11 stars

0 watching

35 forks

Releases 2

EPC 2026 version (Latest) 2 months ago

+ 1 release

Deployments 3

github-pages 4 months ago

Packages

See here:
<https://github.com/akbaritabar/Git-GitHub-Version-Control-Interactive-Self-paced-Course>

The interactive course starts from the basics ...

All issues

is:issue state:closed label:enrollment

☐ Open 0 ☒ Closed 44

☐ [\[Enrollment\] @OpenScienceDemography](#) enrollment
#90 · by akbaritabar was closed on Jul 10

☐ [\[Enrollment\] @Inyan13](#) enrollment
#88 · by akbaritabar was closed on Jul 8

☐ [\[Enrollment\] @MarkNepf](#) enrollment
#86 · by akbaritabar was closed on Jul 1

☐ [\[Enrollment\] @francherrima](#) enrollment
#84 · by akbaritabar was closed on Jun 23

☐ [\[Enrollment\] @geraldinasaraci](#) enrollment
#82 · by akbaritabar was closed on Jun 23

☐ [\[Enrollment\] @giulip00](#) enrollment
#80 · by akbaritabar was closed on Jun 23

☐ [\[Enrollment\] @otorrisi](#) enrollment
#78 · by akbaritabar was closed on Jun 23

☐ [\[Enrollment\] @emmabellotti](#) enrollment
#76 · by akbaritabar was closed on Jun 23

☐ [\[Enrollment\] @eliashofmann](#) enrollment
#74 · by akbaritabar was closed on Jun 23

☐ [\[Enrollment\] @andradelauro](#) enrollment
#72 · by akbaritabar was closed on Jun 23

☐ [\[Enrollment\] @stastnaanna](#) enrollment
#70 · by akbaritabar was closed on Jun 3

@stastnaanna started the Git/GitHub interactive course #71

☒ Closed

 akbaritabar opened on Jun 3

Hi @stastnaanna!

Welcome to the interactive Git and GitHub course.

Participant: stastnaanna

Fork repo: stastnaanna/Git-GitHub-Version-Control-Interactive-Self-paced-Course

Fork sha: ee851a31cd92d4cf0d0e51e6367536ebe5b4d8b3

Enrollment issue: #70

Fork archive: stastnaanna#1

This issue tracks your progress through the course.

Post your answer and the `/done N` command in a comment to advance. Use `/done N` in a new line between `/done` and `N`, where `N` is the step number. The workflow will reply with feedback and then delete the following lines.

 Please keep this issue open. Do not click "Close issue". If you accidentally close it, simply re-open the issue and the course continues from exactly where you left off.

- ☒ 1. Git and version control basics
- ☒ 2. GitHub — remote repositories and online collaboration
- ☒ 3. Branches, collaboration, and open science
- ☒ 4. Advanced Git, GitHub tools, and reproducibility

Create sub-issue

 akbaritabar added newparticipant on Jun 3

Step 1: Why do we need Version Control? What is Git?

This step covers why version control matters and how Git works as a local tool on your computer.

Why use version control?

A common problem in research and collaborative writing is ending up with many files such as:

- paper_final.docx
- paper_final_revised.docx
- paper_final_revised_AA_2026-04-18.docx
- paper_REAL_final.docx

That workflow becomes hard to search, hard to merge, and hard to trust.

In the photo below, you see an example from my own collaborative writing for a published article. We used dates to record the version of the file and find the latest one. We used name initials, i.e., AA is mine, and tracked who did the latest changes. Nonetheless, this is difficult and if you ask me what changed between two versions, I would not know precisely.

Of course, you can think of running a review tab, compare on the two Word files, but how much can you learn from that and how easy will it be to track changes? Will it be easier if you had a short memo or note saying what happened?

-  20240220_Manuscript_mobility_of_talents_LB.docx
-  20240303_Manuscript_mobility_of_talents_LB_AA.docx
-  20240401_Manuscript_mobility_of_talents_2.docx
-  20240401_Manuscript_mobility_of_talents_LB_AA_rh_AA.docx
-  20240408_Manuscript_mobility_of_talents_CHANGES_ACCEPTED_journal_submis...
-  20240511_Manuscript_mobility_of_talents_CHANGES_ACCEPTED_journal_submis...
-  20240515_Manuscript_mobility_of_talents.docx
-  20240521_Manuscript_mobility_of_talents_lb_aa_rh_aa_PREVIOUS_VERSION.docx
-  20240811_Manuscript_mobility_of_talents_REVISSED.docx
-  20240811_Manuscript_mobility_of_talents_REVISSED_lb_aa.docx
-  20241024_Manuscript_mobility_of_talents_alternative_results.docx
-  20241024_Manuscript_mobility_of_talents_alternative_results-lb-aa.docx
-  20241031_Manuscript_mobility_of_talents_alternative_results-lb-aa-rh-aa.docx
-  20241031_Manuscript_mobility_of_talents_alternative_results-lb-aa-rh-aa-lb.docx
-  20241105_Manuscript_mobility_of_talents_alternative_results.docx

Version control replaces that file-name chaos with a tracked and auditable history of changes.

... and covers some more advanced topics.

Reproducible environments — same code, same results

Code that runs on your machine may not run on a collaborator's. Virtual environment tools solve this:

- **Renv** (R): records and restores the exact R package versions used in a project
- **uv** (Python): fast modern Python environment and package manager
 - **Cookiecutter**: In step 1, you learned that having a consistent folder structure is easier to start. **uv** allows you to use a tool called **cookiecutter** which creates project descriptions, according to your wishes.
 - Here is one example that I have set up to create new projects with the same structure as **uv** and **Cookiecutter**: https://github.com/akbaritabar/cookiecutter_project
- **Docker** / **Podman**: containerise the entire computing environment so any machine can run it. On Linux, this is very straightforward, on Windows, you need Windows Subsystem for Linux

NOTE: These will hopefully save you from having to say (or hear): *It works on my computer*

Plain text writing workflows

Git is not only for code. It works equally well for academic writing:

- **Markdown**: lightweight formatting for notes, READMEs, and web content. Even teaching materials were written using markdown syntax that supports what you want. Cool, no?
- **R Markdown** / **Quarto**: combine text, description, and code in one document. You can have a figure/table/results alongside the text that describes them. This has been called a "book" before.
- **LaTeX** / **Overleaf**: professional typesetting with full Git support. You can also use plain text or checkout Quarto presentations!
- **latexdiff**: a Perl script that shows tracked changes between two LaTeX versions. When you send your paper to a journal and need to revise it and show what was changed

Scientific workflows



The image shows a Snakemake workflow report and a diagram of a scientific workflow. The report is titled "Snakemake Report" and shows the workflow "multiple_affiliations_per_author_career_year". The diagram shows a flow from "multiple_affiliations_per_author_career_year" to "multiple_affiliations_per_paper_year", then to "multiple_affiliations_per_region_year", and finally to "multiple_affiliations_per_region_year_by_attribute".

```
workflow > rules > snakefile_map_figures.smk
You, last month | 1 author (You)
1  ## File name to use in search: snakefile_map_figures.smk ##
2
3  # =====
4  ## NMR and MEI MAPS for all time_span combinations ##
5  # =====
6
7  rule plot_map_NMR_AND_MEI_MAPS:
8      input:
9          rules.prepare_data_for_mapping.output
10     output:
11         NMR_AND_MEI_MAPS
12     log:
13         NMR_AND_MEI_MAPS_LOG
14     shell:
15         "(python workflow/scripts/generic_src_mapping_figures.
16         py --input {input} --MEASURE_MAPPED {wildcards.
17         measure_mapped} --MIGRATION_SYSTEM {wildcards.
18         migration_system} --GEO_REGION {wildcards.geo_region}
19         --TIME_SPAN {wildcards.time_span} --output {output})
20         2> {log}"
```

Git aliases

Git aliases — type less, do more

You can shorten long Git commands with **aliases** defined in your `.gitconfig` file:

```
[alias]
st = status
cm = commit -m
ad = add -A
pl = pull --all
ps = push --all
df = diff --color-words
lg = log --pretty=format:'%C(auto)%h%C(blue) %C(19)%s%C(auto)%d %s'
```

Here is a photo showing how a `.gitconfig` file including that alias:

```
home > ali > .gitconfig
18 [alias]
19     #lg = log --oneline --decorate
20     lg = log --pretty=format:'%C(auto)%h%C(blue)
21         %C(19)%s%C(auto)%d %s'
22
23 df = diff --color-words
24
25 st = status
26
27 pl = pull --all
28
29 ps = push --all
30
31 cm = commit -m
32
33 ad = add -A
```

Set one from the command line:

```
git config --global alias.st "status"
```

After that, `git st` runs `git status`. Over time, aliases like these save a lot of typing.

GitHub Pages — publish directly from a repository

RESEARCH

OS best practices;
Reproducible workflows

Open data ==> Enabling future research

Reproducible scientific workflows and reproducibility packages

The screenshot shows a GitHub repository page for 'Internal-and-international-migration-of-scientists' by user 'akbaritabar'. The repository is public and has 2 branches and 1 tag. The file list includes: _publication_figures, config, docs, logs, resources, results, workflow, .gitignore, LICENSE, and README.md. The README section is expanded, showing the title 'Scripts, data, and replication materials for "Global subnational estimates of migration of scientists reveal large disparities in internal and international flows"' and a description: 'This repository includes replication materials, including data, Python, and R scripts to replicate the article's analysis and figures (shown further below) with the following metadata.'

<https://doi.org/10.5281/zenodo.15047102>

The top part of the image shows a PNAS research article titled 'Global subnational estimates of migration of scientists reveal large disparities in internal and international flows' by Aliakbar Akbaritabar, Marce J. Dalko, Xing Zhao, and Emilio Zaghen. The article is in the field of Demography and was accepted for publication on March 4, 2025. A 'Significance' box highlights that migration results in talent circulation and recombination of ideas, and that the study reveals relevant disparities in the attractiveness of subnational regions.

The bottom part shows a Snakefile workflow for generating maps. The workflow is named 'snakefile_map_figures.smk' and is located in the 'workflow' directory. It defines input files (G:/a/scp_rp_2020/scp_rp_2020_authorship_orgs_1st_SOR.parquet, G:/a/SM/resources/migration/internal_mobility_province.parquet, G:/a/SM/resources/migration/international_mobility_province.parquet) and output files (G:/a/SM/results/figures/author_level_mult_affiliations.png, G:/a/SM/results/tables/descriptive_table_multiple_affiliation_hypotypes.tex, G:/a/SM/results/tables/descriptive_table_multiple_affiliation_dominant_type_per_author.tex). The workflow includes a rule 'plot_map_NMR_AND_MEI_MAPS' that uses the 'rules.prepare_data_for_mapping.output' and generates NMR and MEI maps for all time_span combinations. The rule is implemented in Python and R scripts.

My first experience with SnakeMake was great!

Open data ==> Enabling future research

Please note, MPIDR hosts many other cool databases as well... this is only one of them!

THE SCHOLARLY MIGRATION DATABASE

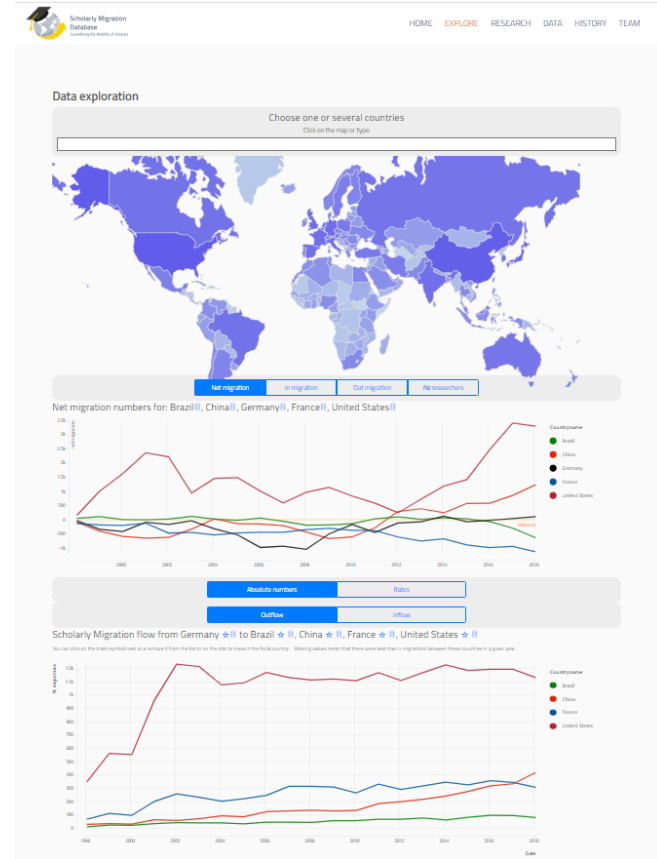


Scholarly Migration
Database
Quantifying the Mobility of Scholars

Scholarly Migration Database Quantifying the Mobility of Scholars

The Scholarly Migration Database provides detailed data about the mobility of scholars. Scientists, journalists and policy analysts who are interested in the movements of published researchers can access high-quality datasets and estimates via this website. The Scholarly Migration Database is prepared and hosted at the Max Planck Institute for Demographic Research in Rostock, Germany.

www.scholarlymigration.org



Subnational level data and dashboard on global migration of scholars



Scholarly Migration
Database
Quantifying the Mobility of Scholars

[HOME](#) [EXPLORE](#) [RESEARCH](#)

Subnational Scholarly Migration Data Explorer

This exploration currently still shows the old data release from 2024. We will update it soon.

Years: 2014 - 2015

Select Data Metric:

Net Migration Rate

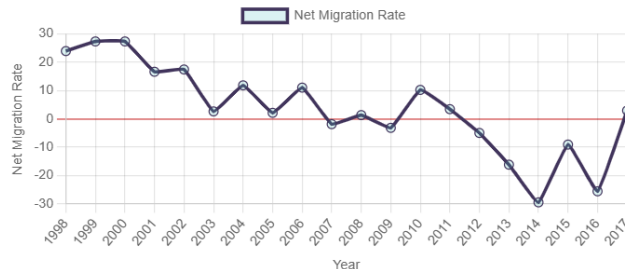
Net Migration Rate - International only

Net Migration Rate - Internal only

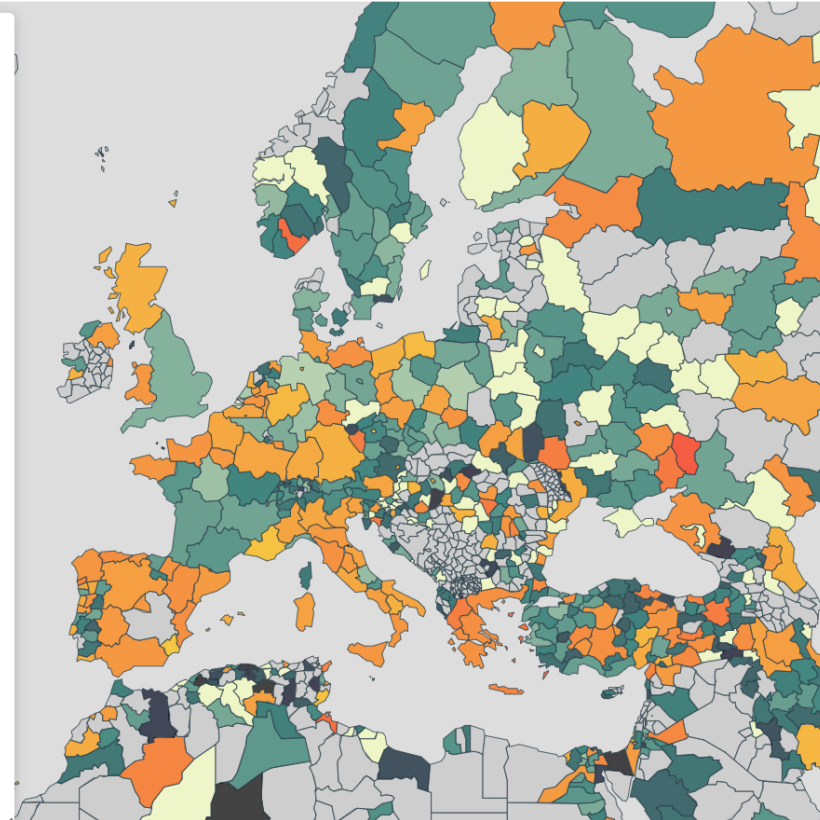
Number of Publishing Scholars

Net migration rate. The sum of the international migration rate and the national migration rate. The measure is shown per 1,000 scholars.

Mecklenburg-Vorpommern (DEU.8_1)



www.scholarlymigration.org



SMD paper and data

www.nature.com/scientificdata

scientific **data**

OPEN

DATA DESCRIPTOR

Bilateral flows and rates of international migration of scholars for 210 countries for the period 1998-2020

Aliakbar Akbaritabar^{1,2}✉, Tom Theile^{1,2} & Emilio Zagheni¹

A lack of comprehensive migration data is a major barrier for understanding the causes and consequences of migration processes, including for specific groups like high-skilled migrants. We leverage large-scale bibliometric data from Scopus and OpenAlex to trace the global movements of scholars. Based on our empirical validations, we develop pre-processing steps and offer best practices for the measurement and identification of migration events. We have prepared a publicly accessible dataset that shows a high level of correlation between the counts of scholars in Scopus and OpenAlex for most countries. Although OpenAlex has more extensive coverage of non-Western countries, the highest correlations with Scopus are observed in Western countries. We share aggregated yearly estimates of international migration rates and of bilateral flows for 210 countries and areas worldwide for the period 1998–2020 and describe the data structure and usage notes. We expect that the publicly shared dataset will enable researchers to further study the causes and the consequences of migration of scholars to forecast the future mobility of academic talent worldwide.

zenodo

Search records...

Communities My dashboard

Log in Sign up

Zenodo is currently experiencing slowness and intermittent outages due to heavy automated traffic from bots and AI crawlers. We are aware of the problem, and our team is focused on stabilizing the service. Thank you for your patience.

Published May 31, 2026 | Version v2

Dataset Open

Replication data for: Bilateral flows and rates of subnational and international migration of scholars worldwide by gender and field

Theile, Tom (Other)¹✉; Akbaritabar, Aliakbar¹✉; Zagheni, Emilio¹✉

Show affiliations

Contributors

Data curator (2): Akbaritabar, Aliakbar¹✉; Theile, Tom¹✉

Data manager: Zagheni, Emilio¹✉

Show affiliations

Data and code for performing analyses and plotting figures for "Bilateral flows and rates of subnational and international migration of scholars worldwide by gender and field"

The code and data can also be found at <https://github.com/MPIDR/Global-flows-and-rates-of-international-migration-of-scholars/>

Abstract: Subnational migration is often more intensive than international migration. These systems of migration are interrelated; however, the literature studies internal and international migration separately. The main reason is that integrated data on both of these migration systems are rare. In the case of scholarly migration, in addition to differences by migration type, trends substantially differ by gender and field of science. We prepare and publicly share the second version of the Scholarly Migration Database (SMD 2.0) including global bilateral flows and rates of subnational and international migration of scholars disaggregated by gender and field of science. We leverage large-scale bibliometric data from Scopus and OpenAlex to construct an integrated history of internal and international movements of scholars at the individual level. We then aggregate these trends to the province level (GeoNames admin 1 regions). Based on our adversarial collaboration and empirical validations using independent and reproducible scientific pipelines, we develop processing steps and offer best practices for the measurement and identification of migration events. We share aggregated yearly estimates of migration rates and of bilateral flows for 2,067 unique subnational regions worldwide in Scopus and 2,080 in OpenAlex for the period 1998–2024. We describe the data structure and provide usage notes. Given the breadth of research using the first version of our database, we expect that the publicly shared second version will further enable researchers to study the causes and the consequences of gender and field differences in the migration of scholars.

Files

Global-flows-and-rates-of-international-migration-of-scholars-2026_V2.zip

1K VIEWS

444 DOWNLOADS

Show more details

	All versions	This version
Views	1,229	367
Downloads	444	123
Data volume	20.3 GB	10.4 GB

More info on how stats are collected...

Versions

Version v2

10.5281/zenodo.20487917

May 31, 2026

Version v1

10.5281/zenodo.11145735

May 8, 2024

View all 2 versions

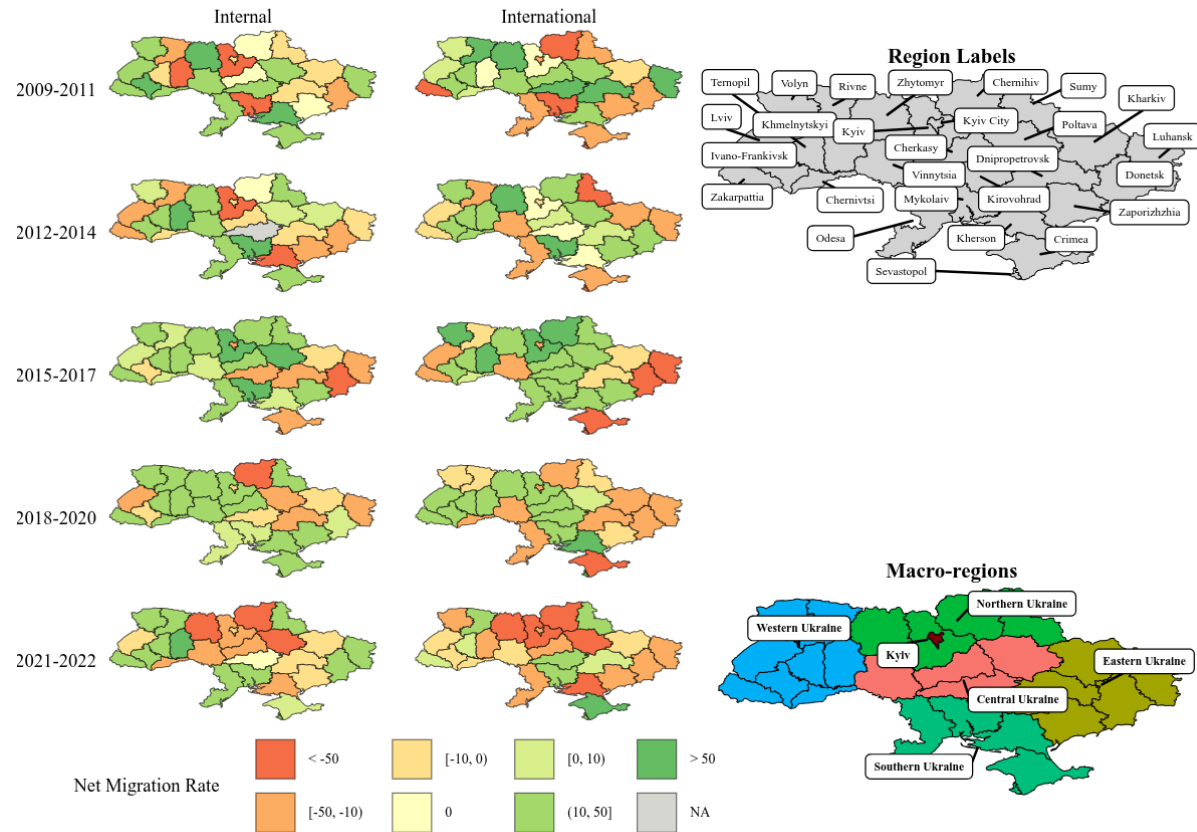
Cite all versions? You can cite all versions by using the DOI [10.5281/zenodo.11145734](https://doi.org/10.5281/zenodo.11145734). This DOI represents all versions, and will always resolve to the latest one. [Read more.](#)

Paper: <https://doi.org/10.1038/s41597-024-03655-9>

Data: <https://doi.org/10.5281/zenodo.11145735>

Example research using SMD data

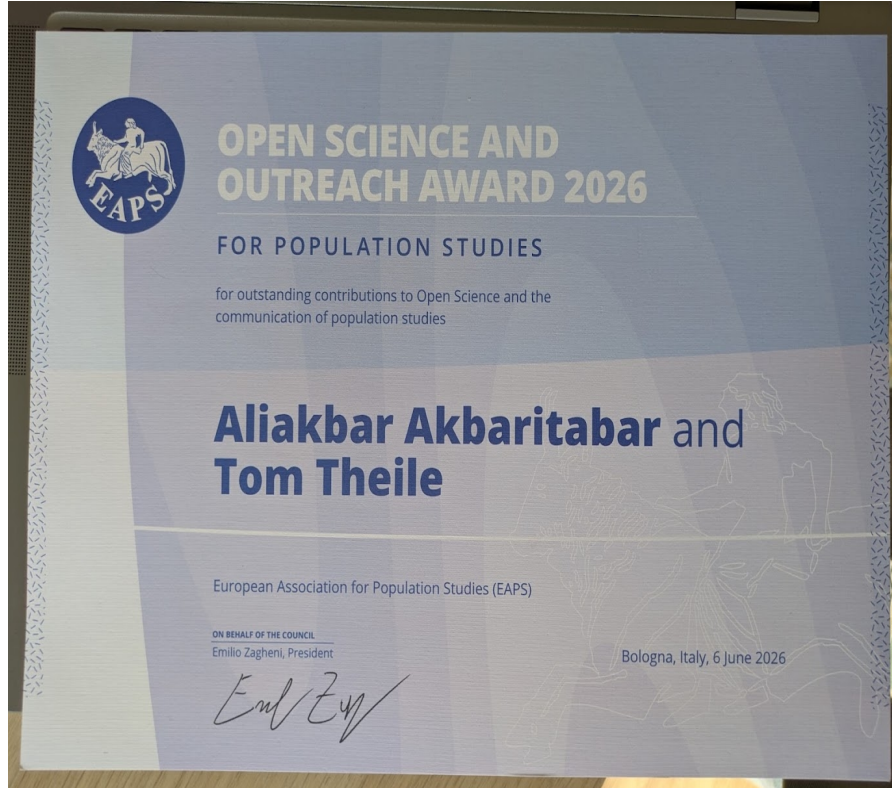
Could external shocks (e.g., conflict) influence in- and out-flow of Ukrainian scholars?



Zaslavsky, Akbaritabar (2026). PDR

<https://doi.org/10.1111/padr.70088>

The European Association for Population Studies has an OS + outreach award!



COMMUNITY

Finding other Open Science enthusiasts

Awesome demographic methods list, started as a list for my students...



Aliakbar Akbaritabar • You
Assistant Professor of Computational Social Science at the Unive...
2mo • 🌐

Prepared a list of R packages, Python libraries, and other tools allowing to implement "demographic methods" for students.

Added a few example reproducibility packages to learn from as well.

If you know of other packages that I should include, please reply or send them as a direct message, thanks 🙏

Course: introduction_to_demographic_methods_2026 / Hands on / 📄

A list of R packages and resources for demographic methods

Please check these R packages and resources on your own. Some could be very helpful for your final essay and make it easier to construct a life table, a Leslie matrix and so on and calculate the measures we discussed during the course.

- Abel, G. J. (2025). *migest*: Tools for Estimating, Measuring and Working with Migration Data (Version 2.0.5) [Computer software]. <https://cran.r-project.org/web/packages/migest/index.html>
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MAKE IT OPEN AND COLLABORATORS WILL JOIN ...

Awesome demographic methods list, community helped the list grow!

[README](#) [Code of conduct](#) [Contributing](#) [CC0-1.0 license](#)

Awesome Demographic Methods

An [awesome](#) curated list of resources for Demographic Methods. Inspired by [Awesome Computational Social Science](#) and others.

The order of entries within categories is either alphabetically or chronologically. **Please add your resources according to the respective ordering**

Contents

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 **Aliakbar Akbaritabar** • You
Assistant Professor of Computational Social Science at the Unive...
1mo • 

Awesome Demographic Methods list is live since a few weeks!

Access and share it with others:
<https://lnkd.in/eVy3cw-j>

Several colleagues generously added their suggestions (Thanks).

You can contribute to it by adding new items here
https://lnkd.in/eaU_4SJE

[awesome-demographic-methods](#)

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Open Science Demography

OpenScienceDemography

OverviewRepositories9DiscussionsProjectsPackagesTeams2People37InsightsSettings

Open Science Demography

For now, you can read our preliminary goals and explore the first version of Awesome Demographic Methods list... more soon!

21 followershttps://github.com/OpenScienceDe...

ReadMe .md

Welcome to Open Science Demography!

Thank you for joining Open Science Demography as a member! It is a pleasure to have you onboard!

Why Open Science Demography?

- **A Dedicated Global Community:** Many population scientists worldwide have spent years contributing to open science, developing open-source software packages (in R, Python, and other languages), sharing open-access teaching materials, building public databases, and advocating for transparent research practices.
- **An Eager New Generation:** A growing cohort of early-career demographers is enthusiastic about open science and looking for ways to participate and contribute.
- **An Opportunity to Connect:** While excellent curated resources like the [Applied Demography Toolbox](#) exist, demographic tools and learning materials are often scattered across personal websites, university portals, and disconnected repositories.

What We Wish to Contribute (very preliminary ideas)

We want to bridge these gaps by building **Open Science Demography** as a global, community-driven space to consolidate resources, share open practices, and foster collaboration across the population sciences.

We aim to:

1. **Consolidate Knowledge:** Maintain a comprehensive, open-access directory of demographic tools, packages, databases, and educational resources.

View as: Member

You are viewing the README and pinned repositories as a member of the Open Science Demography organization.

You can [hide the tasks we've suggested](#) on this page and bring them back later.

Top discussions this past month

Discussions are for sharing announcements, creating conversation in your community, answering questions, and more.

[Start a new discussion](#)

Repositories

awesome-demographic-methods

7 Updated on Jul 18

never-in-union

Forked from RyoMogi/never-in-union

R 0 Updated on Jul 14

github-private

Private

Open Science Demography

The screenshot displays the GitHub repository page for 'OpenScienceDemography'. The navigation bar at the top includes links for Overview, Repositories (9), Discussions, Projects, Packages, Teams (2), People (37), Insights, and Settings. The repository name 'Open Science Demography' is prominently displayed, along with a description: 'For now, you can read our preliminary goals and explore the first version of Awesome'. A yellow box highlights the '21 followers' count. A red box highlights the 'People' section, which shows a grid of 18 member avatars and a 'View all' link. Below the 'People' section is an 'Invite someone' button. The main content area features a 'ReadMe.md' section with a welcome message and a list of reasons to join the community. The 'What We Wish to Contribute (very preliminary ideas)' section outlines the project's goals. On the right sidebar, the 'Repositories' section lists 'awesome-demographic-methods' and 'never-in-union'.

OpenScienceDemography

Overview Repositories 9 Discussions Projects Packages Teams 2 People 37 Insights Settings

Open Science Demography

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- never-in-union** Forked from RyoMogi/never-in-union ● R ☆ 0 Updated on Jul 14
- .github-private** Private

**Where could we take
Open Science Demography?**

What is next in the pipeline?



ABOUT IUSSP MEMBERSHIP SCIENTIFIC GROUPS MEETINGS & EVENTS TRAINING CAREERS RESOURCES PUBLICATIONS

Current Panels

What are IUSSP Scientific Panels?

[Current Panels](#)

[Past Panels and Groups](#)

[Suggest a new Panel](#)

[Data Revolution](#)

[Guidelines for Panels](#)

[Urban Family Planning](#)

[Abortion Research](#) (2019-2026)

[Adult Mortality in Low- and Middle-Income Countries](#) (2026-2030)

[Aligning Population Policies with Low Fertility](#) 2026-2029

[Climate and Environmental Demography](#) 2026-2030

[Climate Change, Gender, and Migration](#) (2026-2029)

[COVID-19, Fertility, and the Family](#) (2023-2027)

[Digital and Computational Demography](#) (2022-2026)

[Epidemics and Contagious Diseases: The Legacy of the Past](#) (2023 - 2027)

[IUSSP Standing Committee on Demographic Terminology](#)

[Kinship Structures, Dynamics and Inequalities](#) (2025-2029)

[Lifetime Migration](#) (2023-2027)

[Mortality Dynamics and Multi-Morbidity at Death](#) (2026-2029)

[Population Environment Research Network \(PERN\)](#)

IUSSP Panel on

“Digital Demography, AI, and Open Science”

2027–2030

Description

The digital revolution continues to reshape demography, both by influencing demographic behaviour and by transforming the data, methods and questions of demographic research. Artificial intelligence, digital platforms and new forms of data create important opportunities, while changing access to data and growing reliance on commercial technologies raise questions about quality, bias, ethics, transparency, reproducibility and sustainability.

Building on the success of the IUSSP Panels on [Big Data and Population Processes](#) (2015–2018), [Digital Demography](#) (2019–2022), and [Digital and Computational Demography](#) (2023–2026), this panel will foster a rigorous, equitable and open international community in digital and computational demography. It will promote responsible uses of AI and digital trace data, critical comparison with established demographic methods and sources, and research practices that remain transparent, sustainable, reproducible and trustworthy as technologies and data infrastructures change.

Three main goals

Wrap up!

- The main areas of **activity**:
 - **Education**: workshops on best Open Science practices. Reproducibility. R Package development. Academy (Software carpentry or rOpenSci for demography?).
 - **Research**: doing research and OS together. Promoting examples/best practices of reproducibility packages.
 - **Community**: connecting people who are enthusiastic about both Demography and Open Science. Open Science Demography. IUSSP Panel on DCD + OS.

What else? suggestions?

Disclaimer: This was by **no means** a comprehensive list of MPIDR's OS activities. These mention only some where I was involved; and to receive your feedback.

Slides

tinyurl.com/OSA26AK

Thank you!

Comments & suggestions are welcome!

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